

From Diagnosis to Solution

SLEEP: Final Report on Implementation and Empirical Validation

Final report — full validation programme including the four-model cross-family matrix

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Professor Guha,

The validation programme is complete — and more complete than the version scoped in July. Beyond the five run groups originally planned, we executed the full experimental matrix on **four models across three families** (Qwen2.5-7B, Qwen2.5-1.5B, Llama-3.1-8B, Mistral-7B): five-seed replication, ten-cycle continual learning, the warm-up extension, and all baselines, identically on each model. Total compute across the programme was roughly 25 GPU-hours ($\approx$ USD \$20). This report gives the numbers, states plainly which claims survived and which did not, and closes with what we believe the paper now contributes.

Three results frame everything. First, the paper’s central findings are now **universal across every model we can test at this scale**: the free-form recall floor sits at $\text{DRA} \approx 0.006$ on all four models while the same models follow in-context facts at up to 0.99, and internal proxy validation falsely confirms 93–100% of consolidations everywhere. Second, our own multi-cycle claim (C6) failed replication in an instructive way: across nine seeded ten-cycle run-pairs, the safety machinery’s preservation plateau is universal in *shape* but varies by three orders of magnitude in *level* across seeds — so the comparison we previously reported is not wrong so much as unstable, and the honest finding is that fixed-horizon single-run comparisons in this field are unreliable. Third, the proposed warm-up repair is a replicated null: sixteen runs, four models, recall never moved.

1. What was run

Run group	Scope executed	Outcome
1. Multi-seed pipeline	5 seeds $\times$ 4 models	Replicated (universal)
2. Ten-cycle continual	9 seeded run-pairs, 4 models	Revised (variance finding)
3. Warm-up extension	16 runs, 4 models, 2 variants	Null (replicated)
4. float32 precision test	7B, original defaults	Replicated (sharpened)
5. Baselines	RAG / in-context / EWC $\times$ 4 models	Revised (surprise)

Every architecture passed a ground-truth verification gate before any experiment ran: three checks asserting that KV injection is bit-identical with an empty bank, effective with a populated one, and exactly reversible when disabled. The gate rejected two defective configurations during the programme — one config-encoding fault and one genuine Qwen-specific assumption in the attention patch — both fixed before a single invalid number was produced. No model failed the repaired gate; Llama’s identity check passed at $\max|\text{diff}| = 0$.

2. Findings that are now universal

2.1 The recognition–recall floor

	Qwen-7B	Qwen-1.5B	Llama-8B	Mistral-7B
Parametric recall (DRA, 5 seeds)	0.006	0.007	0.006	0.007
In-context recall (same facts)	0.760	0.700	0.987	0.927

Every model recalls the facts nearly perfectly when they are visible in context, and at floor when they are stored parametrically through the consolidation pipeline. The dissociation is a property of the mechanism, not of any one model’s pretraining. The weaker-prior hypothesis — that a smaller model would let memory steer generation more easily — is falsified at 1.5B.

2.2 The self-grading discrepancy

Across the four models, the internal surprise-reduction proxy certifies between 4 and 39 consolidations per run (families differ greatly in how many candidates their surprise statistics allocate), yet at most a fraction of one per run survives a direct recall test: false-confirmation rates of 93–100% everywhere. What began as your review item #6 — “add a direct gate” — has become, in our view, one of the paper’s two most durable contributions: proxy self-validation in continual learning can be near-totally wrong, invariantly across architectures.

3. The correction: long-horizon comparisons are unstable

The interim position (single seed, ten cycles) was that naive fine-tuning overtakes SLEEP by cycle ten. Nine seeded run-pairs across the four models show that position was itself under-sampled:

Model	Seed	SLEEP BCP@10	Naive BCP@10	SLEEP DRA	Naive DRA
Qwen-7B	0	4.53	3.36	0.050	0.085
Qwen-7B	1	10.20	6.42	0.032	0.080
Qwen-7B	2	2.71	5.44	0.057	0.078
Qwen-1.5B	0	15.96	21.01	0.045	0.035
Qwen-1.5B	1	230.1	45.83	0.012	0.048
Llama-8B	0	4.83	6.09	0.032	0.083
Llama-8B	1	1530.9	7.11	0.045	0.075
Mistral-7B	0	3.22	2.90	0.008	0.192
Mistral-7B	1	1.64	3.72	0.003	0.160

Three regularities hold in all nine pairs: SLEEP’s BCP always follows a step-then-plateau trajectory (early cycles protected, then a discrete jump to a level that holds flat to cycle ten); naive LoRA’s recall is higher in eight of nine pairs; and no run of either arm keeps BCP < 1.05 past cycle two. What does *not* replicate is the plateau level — 2.7 to 10.2 across three seeds of the identical Qwen-7B configuration, 4.8 versus 1,531 on Llama — three orders of magnitude on unchanged hyperparameters, flipping the cycle-ten “winner” seed to seed (SLEEP 4/9, naive 5/9).

The claim we now make is methodological and, we believe, stronger than the one it replaces: **fixed-horizon, single-run comparisons of continual-learning architectures are unreliable at the horizons and seed counts this literature typically reports.** The clipping mechanism

guarantees a damage ceiling exists; where it lands is effectively a lottery over the optimisation path. We recommend such comparisons state a horizon sweep and a seed spread, or not be made.

4. The warm-up: a replicated null

The pre-registered criterion (DRA 0.10–0.15 at BCP < 1.5) was met nowhere. Across sixteen runs — two variants, two seeds, four models — recall moved by at most ± 0.008 . The 36-parameter gate converges to identity on every model. The LoRA variant demonstrably learns memory-conditioned prediction everywhere (loss falls ~ 1 nat) and lifts recognition where the model survives, yet recall never follows; its preservation cost is family-dependent and, on Llama and Mistral at the untuned learning rate, catastrophic (BCP 10^4 – 10^6). One instructive side-finding: on Mistral the gate learned to *suppress* memory interference it could not learn to exploit, reducing the cost of carrying a populated bank from $8.9\times$ to $2.8\times$.

The diagnosis is thereby strengthened twice over: the recognition–recall gap survived a principled, precedent-grounded repair attempt, on every family, and the repair moved recognition and recall *differentially* — an experimental reproduction of the dissociation itself.

5. Baselines, completed across models

	Qwen-7B	Qwen-1.5B	Llama-8B	Mistral-7B
In-context DRA	0.760	0.700	0.987	0.927
RAG DRA	0.213	0.460	0.520	0.563
EWC-only DRA (BCP)	0.070 (1.12)	0.033 (1.43)	0.073 (1.47)	0.157 (2.53)

The uncomfortable finding stands and is now scoped: at 7B-class scale, EWC alone delivers a better recall-per-preservation ratio than the full SLEEP pipeline (weaker at 1.5B). RAG’s shortfall against in-context is retrieval failure, not comprehension failure.

6. The paper as it now stands

All results above are integrated into the revised paper as primary content — cross-model tables in the results, the nine-run horizon/seed analysis replacing the earlier multi-cycle section, the sixteen-run warm-up table, and the verification-gate methodology in the annexure. The title is unchanged. The contribution set, compressed: a universal recognition–recall dissociation with an in-context control; a universal, near-total proxy-validation failure; a replicated null on the lightweight repair; the precision-floor amendment separated into dtype and magnitude failures; and the long-horizon instability result. The paper no longer contains any headline claim resting on a single model or a single seed.

7. Recommendation and requested input

We consider the empirical programme closed at this scale. The remaining open items are genuinely next-paper material rather than gaps: a many-seed characterisation of the plateau distribution, the warm-up at Memorizing-Transformers training scale with per-family learning rates, and extension to mixture-of-experts and larger models.

We would welcome your judgement on two points: whether the horizon-and-seed instability result is best kept inside this paper (our view: yes, as its methodological anchor) or split into a short companion note aimed at the continual-learning evaluation literature; and the target venue and deadline, so the submission draft can be formatted accordingly.

Respectfully submitted — Aditya Tripathi. Full results (34 machine-readable files), execution logs, implementation, and the revised paper are in the project repository. The proposal (7 July) and interim report (27 July) accompany this document.